or, more simply,

$$\tau_x \sigma_x |sing\rangle = -|sing\rangle.$$

Now this is a bit surprising: |*sing*> is also an eigenvector of $\tau_x\sigma_x$ with eigenvalue −1. It is far less obvious from just looking at |*sing*> that the *x* components of the two spins are always opposite. Nevertheless, every time Alice and Bob measure them, they find that σ_x and τ_x have opposite values. At this point, you will probably not be surprised to learn that the same thing is true for the *y* components.

Exercise 6.6: Assume Charlie has prepared the two spins in the singlet state. This time, Bob measures τ_y and Alice measures σ_x. What is the expectation value of $\sigma_x\ \tau_y$?

What does this say about the correlation between the two measurements?

Exercise 6.7: Next, Charlie prepares the spins in a different state, called $|T_1\rangle$, where

$$|T_1\rangle = \frac{1}{\sqrt{2}}\Big(|ud\rangle + |du\rangle\Big).$$

In these examples, *T* stands for *triplet*. These triplet states are completely different from the states in the coin and die examples. What are the expectation values of the operators $\sigma_z\tau_z$, $\sigma_x\tau_x$, and $\sigma_y\tau_y$?

What a difference a sign can make!